TIC 389751698

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

```
BLS EPHEMERIS
  Period       10.644713 d  (ALIASED, see below)
    -> corrected period: 21.289427 d
  Epoch (t0)   1354.9147 BTJD
  Duration     0.1200 d
  Depth        19.536 %  (195,362 ppm)
  SNR          142.3

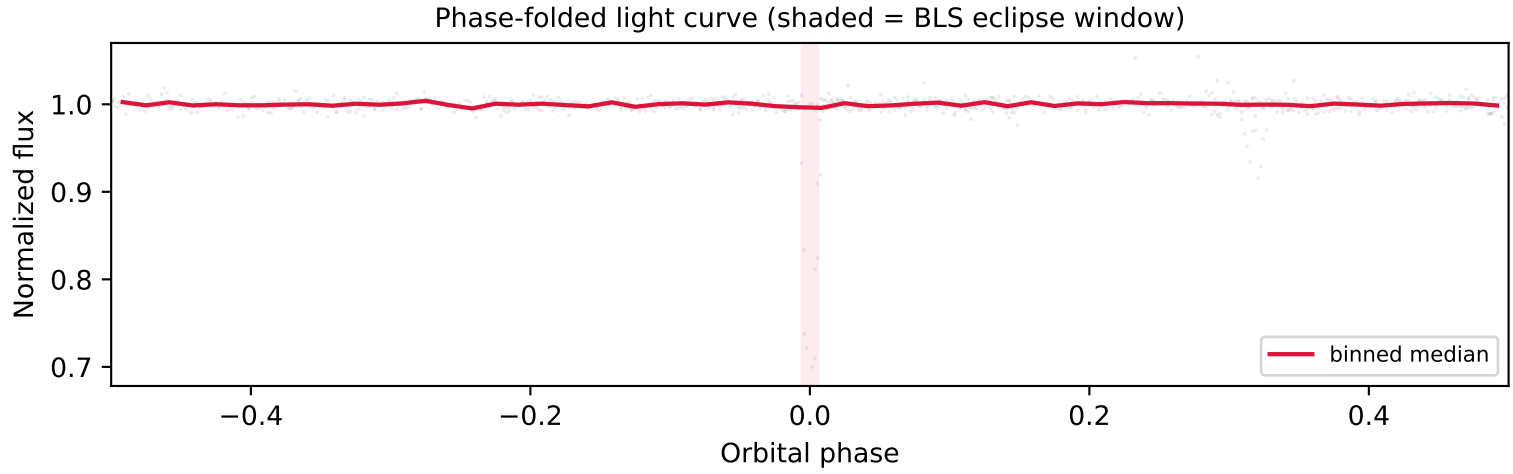
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     3  (gate >= 3)
  Robust depth ratio    0.97  (gate >= 0.50)
  Robust depth sigma    106.7  (gate >= 7.0)
  Odd/even z-score      63.20  (ALIAS FLAGGED)
  Secondary eclipse     not detected
```

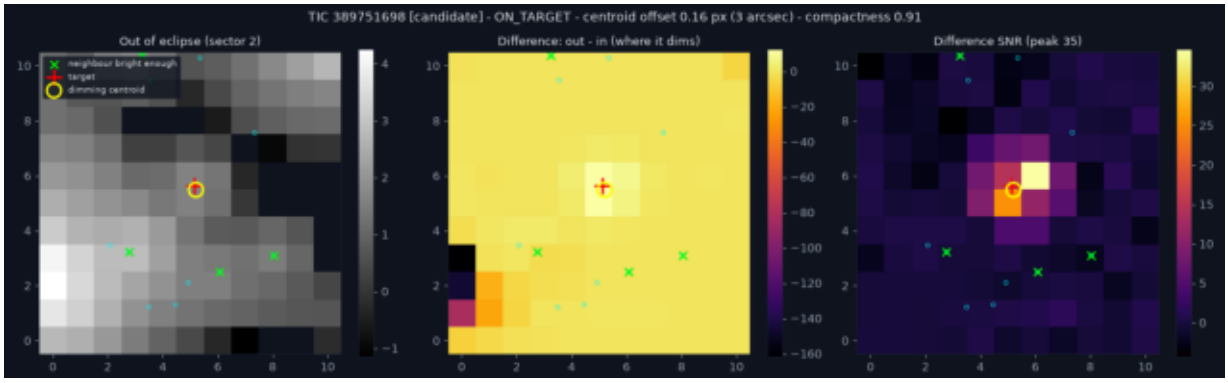
```
CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
```

```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 2    light-curve dip in this sector: 139.2 sigma    t0 refined by 0.0000 phase
  Difference-image peak SNR   34.64   (gate >= 5.0)      Compactness   0.91   (gate >= 0.50)
  Dimming centroid offset     0.16 px (3 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 389728981
    3.11 px away, dTmag +0.55 - farther from the dimming than the target
```





Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 63.20 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 21.289427 d, in any write-up.